

KernelGen Challenge Ceremony

An abstract 3D graphic composed of various geometric shapes, including cubes, prisms, and pyramids, arranged in a complex, layered structure. The shapes are rendered in a light green color with a slight transparency, giving them a floating appearance. They are set against a background of a green-to-yellow gradient and a faint grid pattern.

Bounty Mode Challenge.

3 tasks · 3 leaderboards
\$500 award per track winner

T1: Sparse Attention (Dynamic Sparse Attention)

Implement a Top-K sparse attention kernel that selects KV subsets via gather for attention computation, with an attention sink mechanism.

T2: DeepSeek mHC (Manifold-Constrained Hyper-Connections)

Implement a fused operator combining multi-stream mixing and doubly stochastic matrix projection, completing multi-path residual reads and weighted combination in a single memory access.

T3: Fused MoE (Fused Mixture-of-Experts)

Implement a Mixture-of-Experts fused kernel that merges softmax routing, top-k selection, gate-up GEMM, SiLU gated activation, down GEMM, and weighted reduction into a single call.

test cases from *real-world inference* stages

5 Hardware Platforms

 Ascend
 Iluvatar CoreX
 Hygon
 Moore Threads
 Metax

Timeline

Submission Phase

Start:

May 5, 2026 — 16:00 (Paris Time)
May 5, 2026 — 22:00 (Beijing Time)

Deadline:

May 6, 2026 — 16:00 (Paris Time)
May 6, 2026 — 22:00 (Beijing Time)

Metric

Ranked by **average speedup** across 5 platforms, with Speedup as the core metric



T1: Sparse Attention

yongkang.zou.ai@gmail.com final avg speedup: 1.97

	Huawei Ascend	Iluvatar CoreX	Hygon	Moore Threads	Metax
Speedup for each chip	1.70	3.99	1.11	2.74	0.31

T2: DeepSeek mHC

farid@move37.fr final avg speedup: 80.33

	Huawei Ascend	Iluvatar CoreX	Hygon	Moore Threads	Metax
Speedup for each chip	8.44	55.46	30.12	292.93	14.73

T3: Fused MoE

wqing0093@foxmail.com final avg speedup: 6.16

	Huawei Ascend	Iluvatar CoreX	Hygon	Moore Threads	Metax
Speedup for each chip	5.39	1.74	11.38	4.80	7.50

